BDI Agent Architectures

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References & Sources

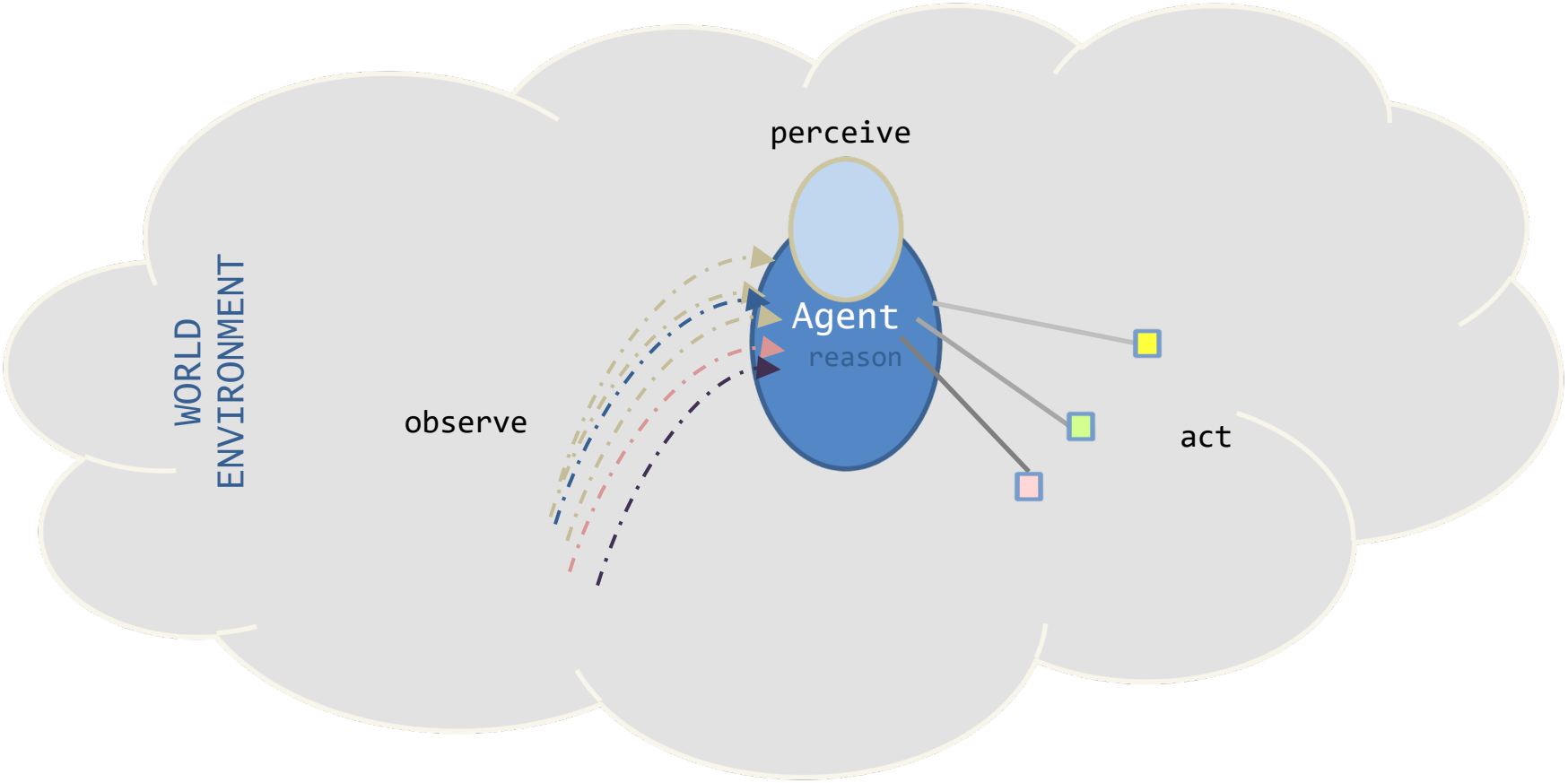
General references

- Bordini, R.H., El Fallah Seghrouchni, A., Hindriks, K., Logan, B. and Ricci, A., 2020. Agent programming in the cognitive era. *Autonomous Agents and Multi-Agent Systems*, *34*(2), p.37
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Specific probabilistic extensions

- Price, C.S., Moodley, D., Pillay, A.W. and Rens, G.B., 2022. An adaptive, probabilistic, cognitive agent architecture for modelling sugarcane growers' operational decision-making. South African Computer Journal, 34(1), pp.152-191.
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The agent paradigm



The BDI agent architecture

Introduced in Bratman's practical reasoning system (Bratman et al 1988) - based on the mental attitudes of *beliefs*, *desires* & *intentions.*

Represents three key mental states of the agent

- Beliefs - what it believes about its environment and itself
- Desires - the set of goals it wants to achieve
- Intentions - goals it is currently committed to achieving

The BDI architecture formed the basis for much of the research on architectures for autonomous agents over the last 30 years

BDI – practical reasoning

Two steps in practical reasoning

1. Deliberation - decide what state of affairs to bring about from the (possibly conflicting) desires of the agent

2. Means-ends reasoning - decide how to bring about that state of affairs.

BDI – practical reasoning

The BDI Cognitive Loop

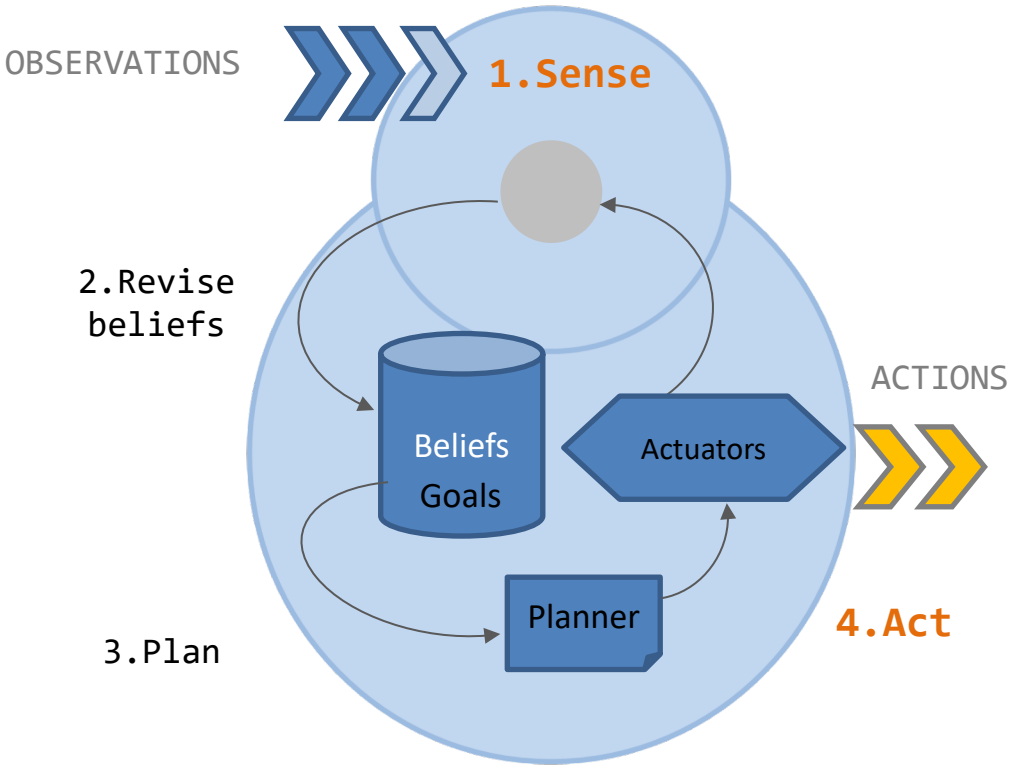
- Receives percepts & revises what it believes about the world and itself
- Deliberates about its goals and current intentions, considers its options & selects a goal to follow, which becomes its intention
- It focuses on the most suitable intentions, finds the most suitable plan & executes it

Using the BDI architecture, a course of action can be selected based on limited information, in a bounded rational way (Bratman et al., 1988).

Without bounding, a rational agent would explore all the options and expect all information at each time step. This mimics how humans take decisions, using “satisficing” (Simon, 1996).

The BDI agent architecture

- **Beliefs** – current state of the environment & itself (the context), and inference rules for reasoning (knowledge)
- **Desires** – overall goals or objectives of the agent, what the agent wants to achieve in the future - can be inconsistent, unrealistic, may conflict
- **Intentions** – current subset of active desires/goals, must be consistent & achievable in the current context



Original BDI control loop

Algorithm 1 BDI agent's control loop (Wooldridge, 2000)

```
1: B := B_0;                          ▷ B_0 are the initial beliefs
2: I := I_0;                          ▷ I_0 are the initial intentions
3: while true do
4:     get next percept ρ;
5:     B := brf(B, ρ);
6:     D := options(B, I);
7:     I := filter(B, D, I);
8:     π := plan(B, I);
9:     execute(π);
10: end while
```

Abstract BDI control loop

Algorithm 1 BDI Agent Interpreter

```
1: procedure AGENTINTERPRETER(⟨Ev, Bel, PLib, Int⟩)
2:     while true do
3:         Ev ← UPDATEEVENTS(Ev)
4:         Bel ← UPDATEBELIEFS(Ev, Bel)
5:         Int ← SELECTPLANS(Ev, Bel, PLib, Int)
6:         Ev ← EXECUTEINTENTION(Int, Ev)
```

From: De Silva, L., Meneguzzi, F. and Logan, B., 2020. BDI agent architectures: A survey. International Joint Conferences on Artificial Intelligence.

Implementing BDI agents

Agent Oriented Programming Languages

- AgentSpeak
  - JASON implements AgentSpeak
  - ASTRA based on AgentSpeak
- PRS – Procedural Reasoning System
- JADEX – programming agents in Java

Advantages of BDI agents

BDI-based agent programming languages offer significant advantages for developing autonomous systems

- rapid development
- context sensitive
- robust behaviour
- greater intelligibility
- verifiability compared to agents where programs are learned

Extending the BDI Architecture

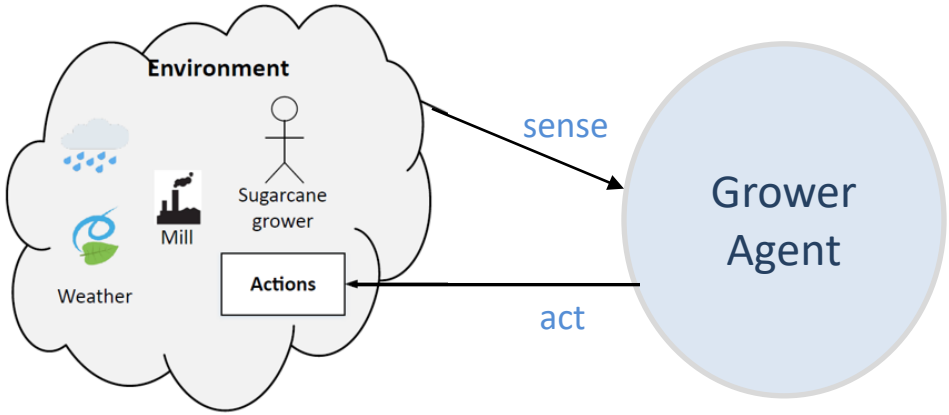
- Goals as hierarchical plans
- Reconsider function if the environment is changing very fast
    - comes just before options(.) and filter(.),
    - decides whether to consider new intentions,
    - saves computation/time,
    - keeps agent focused on current objectives.
- At every iteration through main loop, test whether currently pursued intentions are still achievable, or already achieved (by serendipity)
- Incorporating uncertainty, e.g. POMDP or DDNs

Example - sugarcane grower simulation

Based on a real-world sugar cane supply chain in KwaZulu-Natal

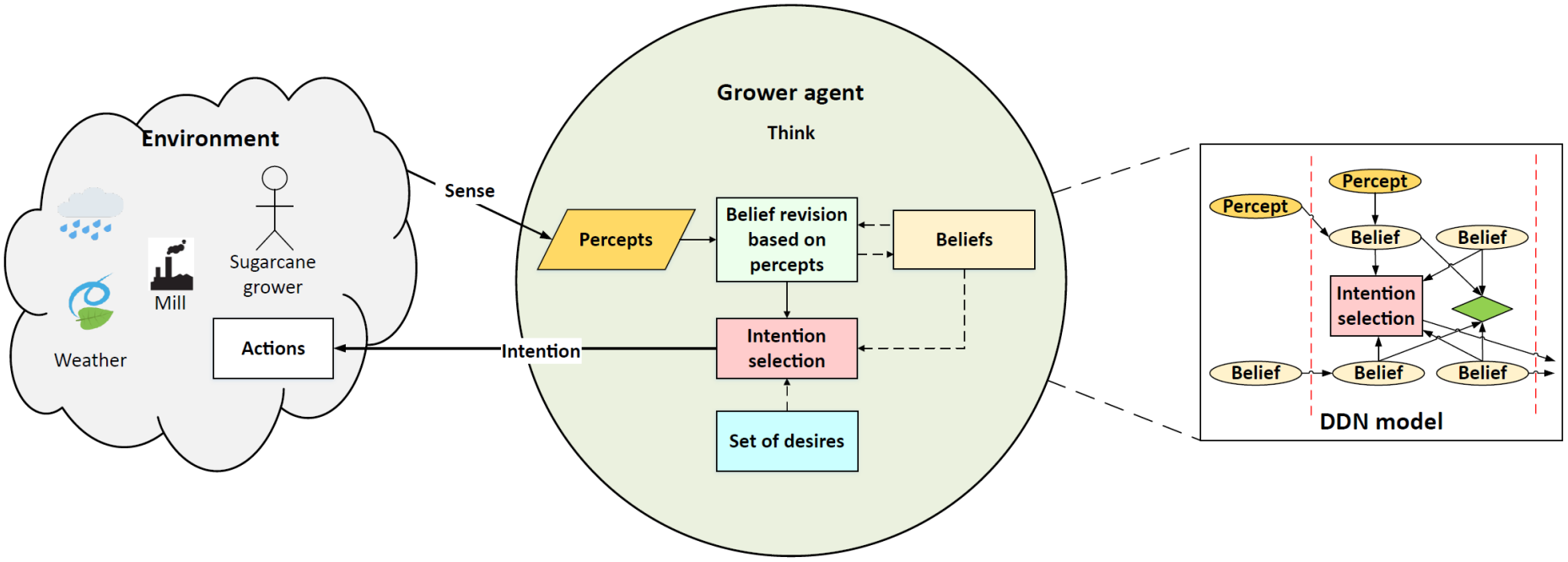
One of the key decision taken by growers (farmers) is when to cut sugar cane crop to deliver it to the mill

- The cane must be burned before delivery
- Burning can not happen when there is rain or high wind.
- The grower has a quota for weekly delivery to the mill

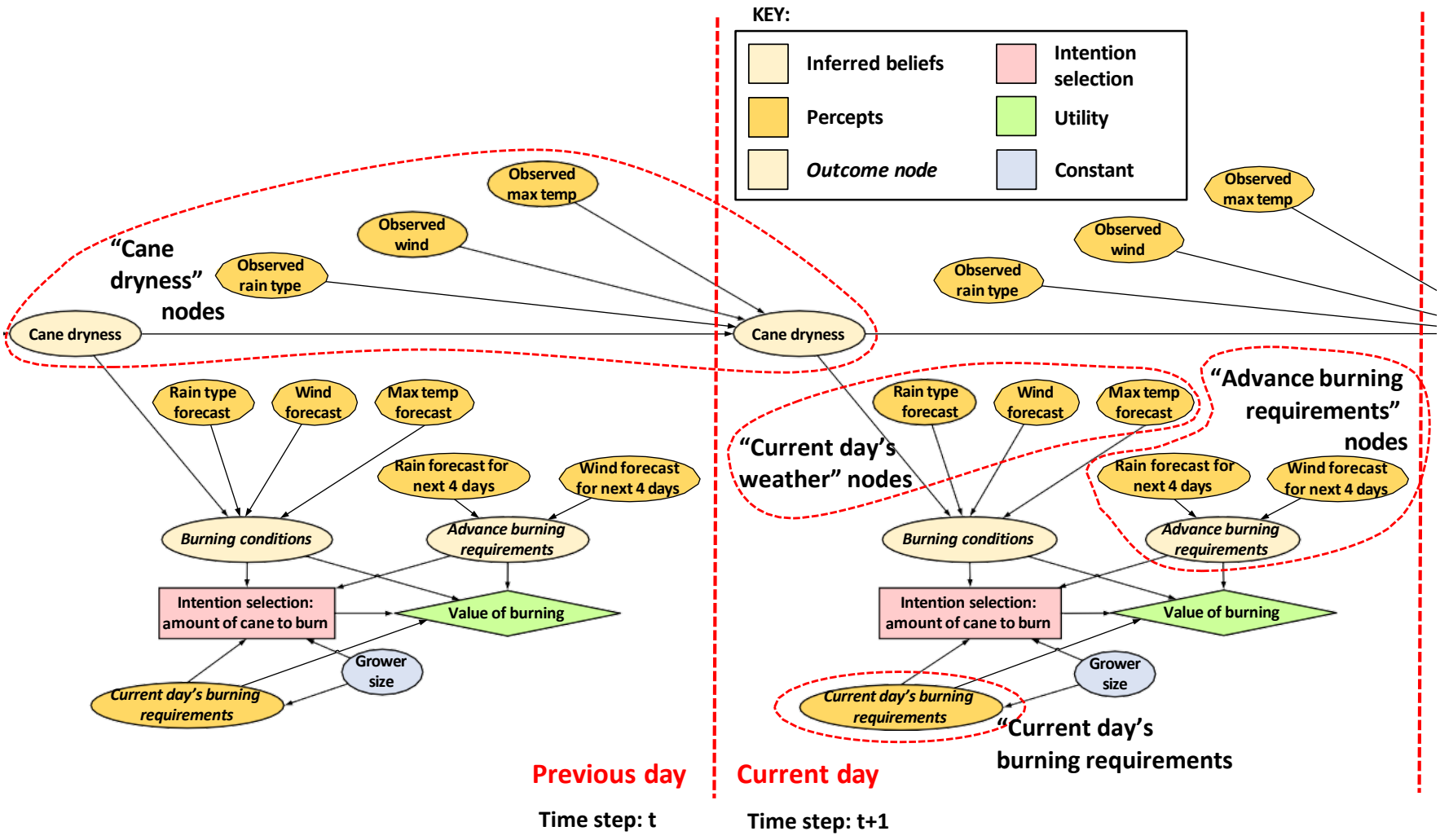


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Probabilistic BDI Agent



Dynamic Bayesian Decision Network



Probabilistic BDI control loop

Algorithm Probabilistic BDI agent's control loop, adapted from Wooldridge (2000)

1: **Input**: DDN, containing initial beliefs and desires relating to cane burning;

2: **Output**: Selected intention
3: $t = 0$ ▷ **time**
4: B_0 = Beliefs stored in the DDN
5: D = set of desires
6: $I_0 = \emptyset$
7: set GrowerSize (large/small) and MaxDailyDeliveryRate
8: **while** true **do**
9: $t = t + 1$
10: get observations from environment
11: P_t = discretised observations ▷ **percepts**
12: $\text{BurningRequirements}_t$ = calcBR($\text{CaneAwaitingDelivery}_t$, DeliveryStatus_t, MaxDailyDeliveryRate, GrowerSize)
13: enter relevant percepts P_t into DDN as evidence
14: B_t = revised beliefs, using Bayesian inference
15: **if** $\text{EmergencyIsPresent}_t$ **then**
I_t = address emergency
enter "Burn no cane" into DDN decision node as evidence
16: **else if** DayOfWeek_t = Sunday **then**
I_t = rest
enter "Burn no cane" into DDN decision node as evidence
17: **else if** it is MillingSeason **then**
I_t = DDN decision state with highest utility
18: **end if**
19: output I_t
20: **end while**

Some discussion points

- Do BDI agents *think* or *act rationally* or *humanly*?
- Are they deliberative, reactive or hybrid?
- How do BDI agents compare to purely ML based learning agents, e.g. Deep Reinforcement Learning agents?

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